Setting up the R package knitr to integrate R & LaTeX

(version 4.4)

Jose Barrera[1,2]

[1]ISGlobal Barcelona Institute for Global Health - Campus MAR
[2]Departament de Matemàtiques, Universitat Autònoma de Barcelona

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E-mail address: jose.barrera@uab.cat
Personal website: `https://jobago.github.io`

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Warning: All screenshot images shown in this document correspond to RStudio (with customized appearance) on Ubuntu 24.04.4 LTS. The appearance can vary in other versions.

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1 Installation

You need to install the following software, in the order indicated.

1.1 R

First, you need to get R installed. **Last version is strongly recommended**. Then, you need to get the R package knitr installed:

```
> install.packages("knitr", dependencies = TRUE)
```

1.2 LaTeX

Now, you need to get a LaTeX editor installed. The installation depends on your computer operative system. When possible[1], a full installation is recommended.

According to my experience with students, the following options seem to be the most likely to work (if you have another option already installed, you can try with that):

- **Linux:** TeXLive (`http://tug.org/texlive/`):
  - In Red Hat (Fedora, CentOS, etc.):

    ```
    sudo dnf install texlive-scheme-full
    ```

  - In Debian/Ubuntu:

    ```
    sudo apt install texlive-full
    ```

- **Windows:** MiKTeX (`https://miktex.org/`).
- **Mac OS:** MacTeX (`http://tug.org/mactex/`).

Alternatively, you can configure RStudio to work with `tinytex`. `tinytex` is a lightweight, portable LaTeX distribution that starts with a minimal size (around 150 MB) and automatically downloads and installs any missing LaTeX packages on the fly whenever you compile a document. See Section 1.3 to configure `tinytex`.

1.3 RStudio

1. Installation: `https://posit.co/download/rstudio-desktop/` **(last version is strongly recommended)**.
2. Setting up (see Figure 1):

> **CRUCIAL STEP:** By default, RStudio uses the old `Sweave` engine. You **must** switch it to knitr before compiling:
>
> a) In the menu, go to Tools → Global Options... → Sweave.
>
> b) Set "Weave Rnw files using:" to **knitr**.
>
> c) Set "Typeset LaTeX into PDF using:" to **pdfLaTeX**.

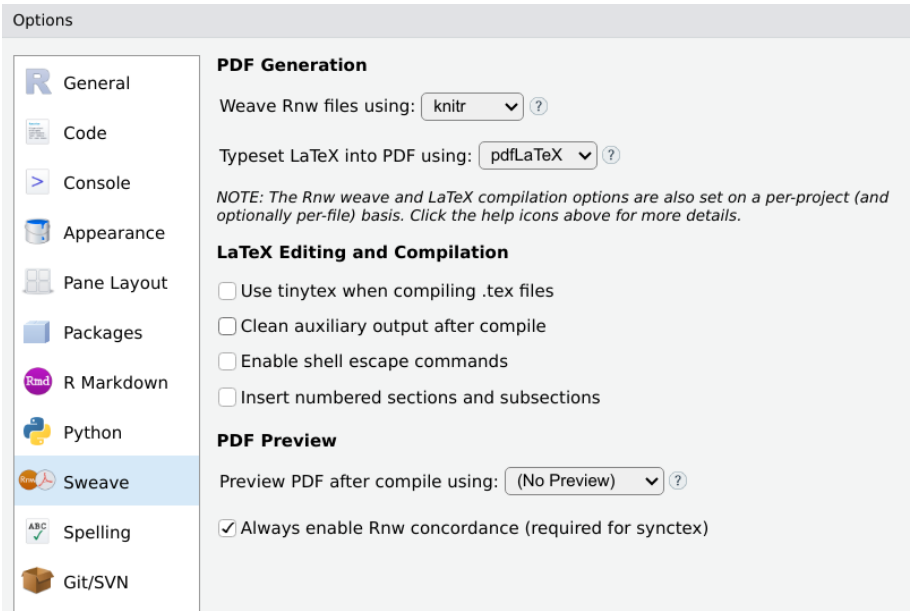


Figure 1: Setting up RStudio for "Sweaving" with `knitr` and LaTeX.

You can set `tinytex` as the LaTeX compiler in RStudio. To install and set up `tinytex`, run the following code in your R console:

```
> install.packages("tinytex")
> tinytex::install_tinytex()
```

and then check the box "Use tinytex when compiling .tex files" in Tools → Global Options... → Sweave (see Figure 1).

Doing this helps avoid path configuration and missing package issues when compiling your `Rnw` documents with knitr. If you experience any issues regarding `tinytex`, refer to its official documentation at `https://yihui.org/tinytex/`.

2 Literate programming with R and LaTeX using the R package `knitr`

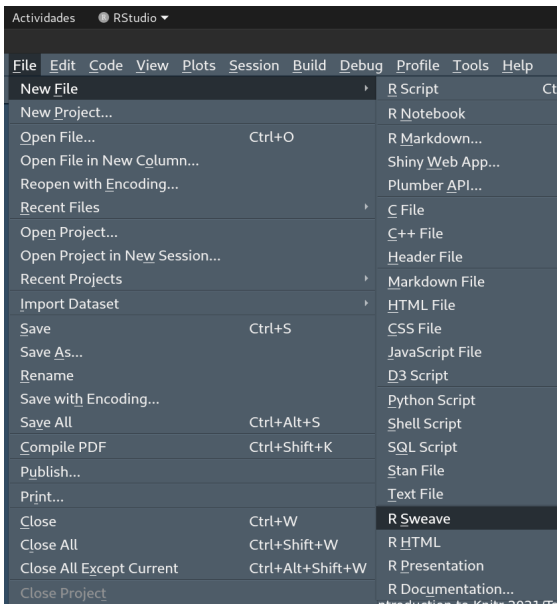
The knitr package was designed to be a transparent engine for dynamic report generation with R. Essentially, it allows to interactively handle the R script and the documentation (i.e. LaTeX code) in a single source `Rnw` file[2]. Essentially, the knitr package works in two steps. In the first step, R internally executes all the R code included in the `Rnw`, and translated the resulting output to LaTeX code, resulting in a new `tex` file. In the second step, LaTeX compiles the `tex` file to get the final `pdf` document. In RStudio, this two-step process can be automatically done by a simple 1-clic compilation, as we explain next.

[1] Taking into account storage space requirements.

[2] Translation of `Rnw` files that have been writen using (obsolete) R function `Sweave` to knitr syntax is straightforward with the `Sweave2knitr` function in the knitr package. The knitr package is more powerful and flexible than the `Sweave` function. In fact, the knitr package can be thought of as an extension of `Sweave`. Visit `http://yihui.name/knitr/` for further information.

3 Compilation

1. Open a session in RStudio.
2. Open or create your `Rnw` file. For instance, open a new, empty document (see Figure 2 (left)) and write the code shown in Figure 2 (right).




```
1  \documentclass[a4paper,10pt]{article}
2  \usepackage[utf8]{inputenc}
3
4  \title{My title}
5  \author{My name}
6  \date{\today}
7
8  \begin{document}
9
10 \maketitle
11
12 Hello.
13
14 <<>>=
15 x <- 2
16 y <- 2 * x
17 y
18 @
19
20 Bye.
21
22 \end{document}
```

Figure 2: Left: How to open a new `Rnw` file in RStudio. **Right:** Content of the `example.Rnw` document.

3. Save the document with the name `example.Rnw`, in your working folder. **In order to make the document compatible with other operating systems, it is very important that you save it using UTF-8 encoding (check it using File → Save with Encoding...).**
4. To compile the `Rnw` file to get the final `PDF`, you have two ways:
   - A **One-click action**, which is **fully automatic, easy and fast**, **but offers less error detail when compilation fails**.
   - A **Two-step procedure**, which is **more manual and slower** but **provides detailed error logs** when troubleshooting issues.

3.1 One-click compilation

For one-click compilation, click on the Compile PDF button as shown in Figure 3 (left) or use the keyboard shortcut (`Ctrl + Shift + K` on Windows/Linux, `Cmd + Shift + K` on macOS).

If no errors occur, `example.tex` and `example.pdf` (in addition to other auxiliary files) will be generated in your working folder. Then, open the file `example.pdf` and check that its content is similar to that shown in Figure 3 (right).

3.2 Two-step compilation

When one-click compilation fails with cryptic or abbreviated messages, switching to two-step compilation helps isolate whether the issue lies in the R code or in the LaTeX typesetting.

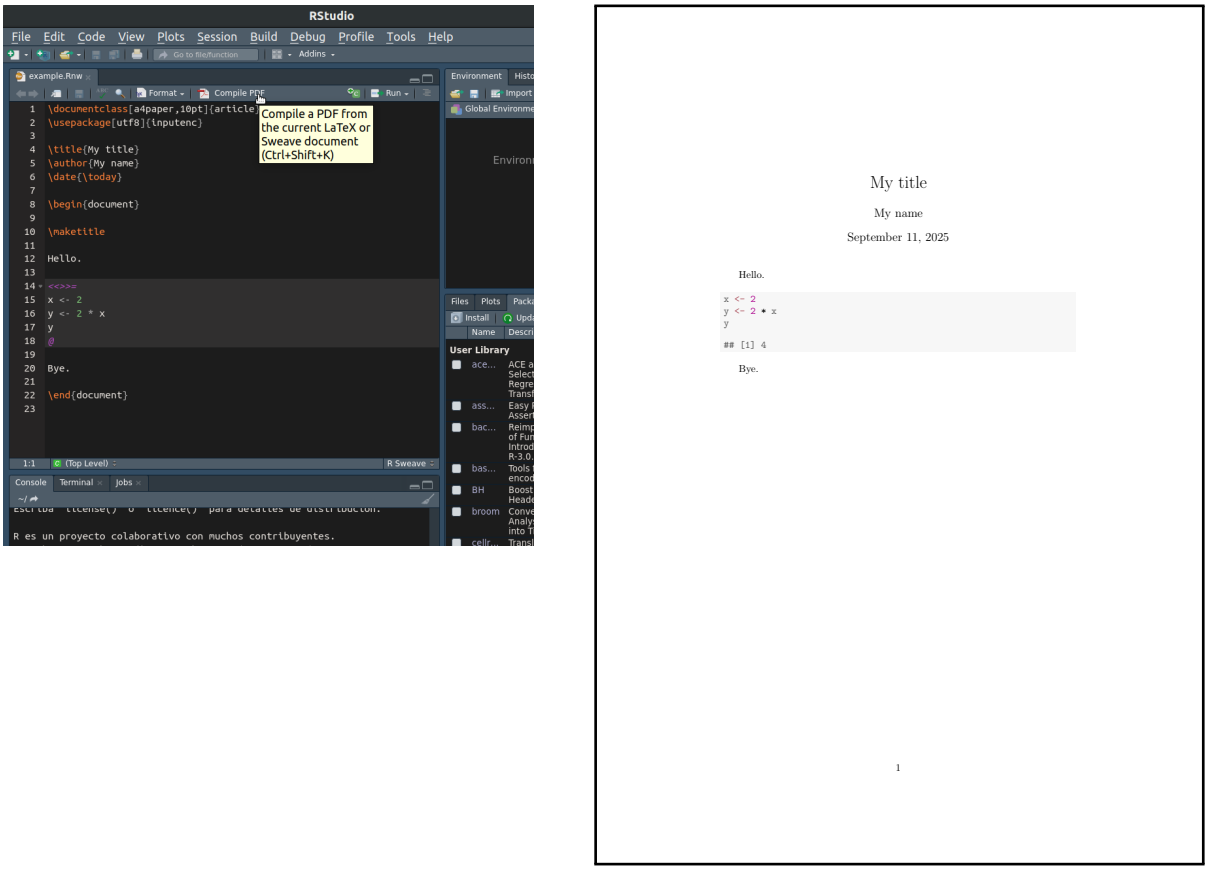


Figure 3: One-click compilation in RStudio. **Left:** Compiling button acts on `example.Rnw`. **Right:** Appearance of the resulting `example.pdf` document.

1. Open an RStudio session and set the folder where you have the `example.Rnw` file as the working directory.

2. In the R console, write and execute:

```
> library(knitr)
> knit("example.Rnw")
```

Then, if no errors occur in the R code execution, the console should show a message similar to the following:

```
> library(knitr)
> knit("example.Rnw")


processing file: example.Rnw
  |.....................                                  |  33%
  ordinary text without R code

  |.........................................              |  67%
label: unnamed-chunk-1
```

```
  |.....................................................| 100%
ordinary text without R code


output file: example.tex

[1] "example.tex"
>
```

3. Check that the file `example.tex` has been created in your working folder.
4. Open the file `example.tex` with your LaTeX compiler (or in RStudio) and compile it.
5. If LaTeX fails during this step, the compiler output or log file will show the exact line number and specific error message, making it much easier to identify and fix the problem.
6. Check that the file `example.pdf` has been created in your working folder and verify its content.

4 Writing in Catalan or Spanish (Special Characters and Accents)

When writing documents in Catalan, Spanish, or other languages that use non-ASCII characters (such as à, é, ç, ñ, ü), please follow these recommendations:

- **File encoding (crucial):** To ensure seamless compatibility when sharing `Rnw` files across different operating systems (Windows, macOS, and Linux[3]), **always save your files using UTF-8 encoding**. In RStudio, go to Tools → Global Options → Code → Saving and set Default text encoding to **UTF-8**. If you ever notice distorted characters when opening a document, go to File → Reopen with Encoding... and select **UTF-8**.
- **Language packages:** If you write in a language other than English, use the babel package to automatically translate document labels such as "Table", "Figure", or "Contents". For instance, adding:

  ```
  \usepackage[catalan]{babel}
  ```

  or

  ```
  \usepackage[spanish]{babel}
  ```

  in the preamble of your `Rnw` file will automatically replace "Table" with "Taula" or "Tabla" in the final `PDF`.

5 Resources

- knitr web page: `https://yihui.org/knitr/`
- Book "Dynamic Documents with R and knitr, 2nd edition": `https://www.taylorfrancis.com/books/mono/10.1201/9781315382487/dynamic-documents-knitr-yihui-xie`

[3] The instructor works on Ubuntu Linux and will grade your work in this environment.